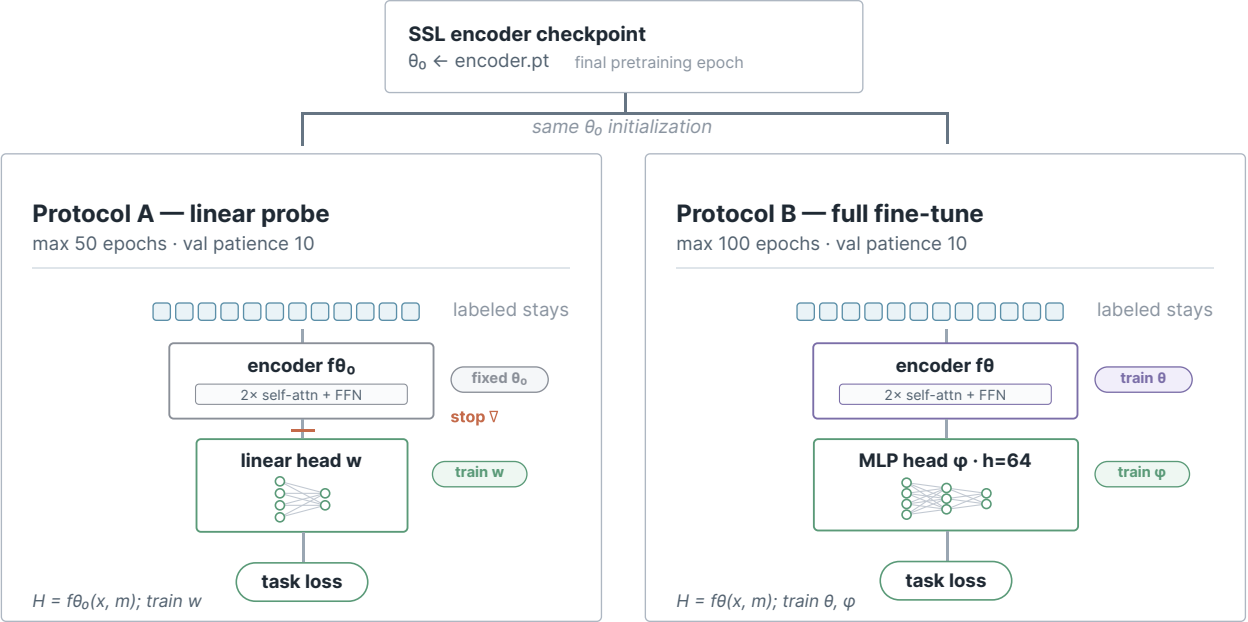
SSL encoder checkpoint
θ₀ ← encoder.pt
final pretraining epoch
same θ₀ initialization
Protocol A — linear probe
max 50 epochs · val patience 10
labeled stays
encoder fθ₀
2× self-attn + FFN
fixed θ₀
stop ∇
linear head w
train w
task loss
H = fθ₀(x, m); train w
Protocol B — full fine-tune
max 100 epochs · val patience 10
labeled stays
encoder fθ
2× self-attn + FFN
train θ
MLP head φ · h=64
train φ
task loss
H = fθ(x, m); train θ, φ